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Blood Hormones as Markers of Training Stress and Overtraining

Axel Urhausen, Holger Gabriel and *Wilfried Kindermann*

Institute of Sports and Preventive Medicine, University of Saarland, Saarbrücken, Germany

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Summary

An imbalance between the overall strain experienced during exercise training and the athlete's tolerance of such effort may induce overreaching or overtraining syndrome. Overtraining syndrome is characterised by diminished sport-specific

physical performance, accelerated fatiguability and subjective symptoms of stress. Overtraining is feared by athletes yet there is a lack of objective parameters suitable for its diagnosis and prevention.

In addition to the determination of substrates (e.g. lactate, ammonia and urea) and enzymes (e.g. creatine kinase), the possibilities for monitoring of training by measuring hormonal levels in blood are currently being investigated.

Endogenous hormones are essential for physiological reactions and adaptations during physical work and influence the recovery phase after exercise by modulating anabolic and catabolic processes. Testosterone and cortisol are playing a significant role in metabolism of protein as well as carbohydrate metabolism. Both are competitive agonists at the receptor level of muscular cells. The testosterone/cortisol ratio is used as an indication of the anabolic/catabolic balance. This ratio decreases in relation to the intensity and duration of physical exercise, as well as during periods of intense training or repetitive competition, and can be reversed by regenerative measures. Correlations have been noted with the training-induced changes of strength. However, it seems more likely that the testosterone/cortisol ratio indicates the actual physiological strain in training, rather than overtraining syndrome.

The sympatho-adrenergic system might be involved in the pathogenesis of overtraining. Overtraining appears as a disturbed autonomic regulation, which in its parasympathicotonic form shows a diminished maximal secretion of catecholamines, combined with an impaired full mobilisation of anaerobic lactic reserves. This is supposed to lead to decreased maximal blood lactate levels and maximal performance. Free plasma adrenaline (epinephrine) and noradrenaline (norepinephrine) may provide additional information for the monitoring of endurance training. While prolonged aerobic exercise conducted at intensities below the individual anaerobic threshold lead to a moderate rise of sympathetic activity, workloads exceeding this threshold are characterised by a disproportionate increase in the levels of catecholamines. In addition, psychological stress during competitive events is characterised by a higher catecholamines to lactate ratio in comparison with training exercise sessions. Thus, the frequency of training sessions with higher anaerobic lactic demands or of competition, should be carefully limited in order to prevent overtraining syndrome.

In the state of overtraining syndrome and overreaching, respectively, an intraindividually decreased maximum rise of pituitary hormones (corticotrophin, growth hormone), cortisol and insulin has been found after a standardised exhaustive exercise test performed with an intensity of 10% above the individual anaerobic threshold. This disturbed stress-response corresponds to findings with insulin-induced hypoglycaemia in overtraining suggesting an impaired hypothalamic regulation.

However, the role of hormones in the recovery phase and their effect on the receptor and intracellular level remain to be better established. Reference values indicating a 'normal' exercise tolerance as well as easier and less expensive laboratory methods are still lacking. External factors influencing the hormonal blood levels require well-standardised sampling conditions which are often difficult to realise in the training environment. The impaired exercise-induced maximal increase of selected hormones and the potential consideration of the psychological stress component by hormonal measurements, however, represent interesting basic findings which encourage future investigations.

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The efficiency of physical training essentially depends on intensity, volume and periodisation (i.e. the organisation of exercise and recovery periods) of the training stimuli.[1] Initially, a catabolic phase with decreased tolerance of effort, characterised by reversible biochemical, hormonal and immunological changes, exists after each exercise session. An anabolic phase follows, with a higher adaptive capacity and enhanced performance capacity. During this anabolic stage the training stimulus is most effective (supercompensation). Therefore, an optimal training programme needs an adequate assessment of the individual's current tolerance of stress. Apart from training and competition, however, additional stressors resulting from diseases and injuries, as well as the professional and social areas of the athlete's life, are also to be taken into consideration.

An imbalance between the overall strain of training and the individual's tolerance of stress leads to overtraining.[1-3] Short term overtraining is called overreaching, and can be reversed by a more prolonged period of recovery. Further exposure to stressors will induce overtraining syndrome or 'staleness' despite a prolonged period of rest e.g. >2 weeks.[4] The more advanced so-called parasympathicotonic ('addisonoid') form of overtraining syndrome is typically associated with less obvious symptoms, while the predominantly sympathicotonic ('basedowoid') form of the overtraining syndrome is easier to recognise because of more pronounced indicators.[5,6] A mixture of these two forms often exists, because the sympathicotonic type can represent an early transitory stage that shifts to parasympathicotonic after individually varied periods of time. Generally overtraining is characterised by the following symptoms: a decreased sports-specific performance capacity when the training load is maintained or even increased; an enhanced fatiguability; and more or less pronounced vegetative complaints, such as altered psychic excitability, sleep disorders, emotional instability, and organ-related complaints which are without organic disease.[5,6] Overtraining has to be differentiated from the 'banal' short term (lasting maximally 1 to 2 days) fatigue after overload training,[2] sometimes also called 'overtraining' which causes confusion with the 'overtraining syndrome'.

1. Overtraining: a Diagnostic Problem

The frequent occurrence of overtraining contrasts with available diagnostic tools. A number of parameters of overtraining are claimed to be identified,[2,3,7] but the data are frequently less based on well-founded experimental findings than on experience or individual observations. Current paraphrases of this state like a 'vegetative dysregulation',[6] (overtraining is not a 'disease'), are still almost identical with the earliest data in the literature: Herxheimer was the first to draw attention to a 'regulation shift of the vegetative nervous system'.[8]

1.1 Exercise Testing in Overtrained Athletes

The evaluation of the decrease in performance as the cardinal symptom during overtraining often causes difficulties. It requires a specific and standardised test methodology and knowledge of the mechanisms of energy supply and performance limiting during overtraining. Furthermore it is doubtful whether results e.g. those obtained by cycle ergometry, can be used for athletes involved in a sport discipline which involves running. However, it can be difficult to design tests that diagnose specific performance under laboratory conditions for some sports, e.g. ball games, racquet sports, etc. The few existing findings indicate an impairment of speed endurance speed endurance or short term endurance with reduced maximal anaerobic lactic capacity.[6,9-11] The decreased performance is primarily recognised when the limit of physical performance is reached (exercise to exhaustion). Testing for overtraining using performance indicators would require setting the desired intensity at standardised exercise protocols above the maximal lactate steady state, in order to guarantee sufficiently high metabolic and psychological strain. During endurance exercise below the range of the individual anaerobic threshold and subsequent longer ex-

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ercise duration, the anaerobic energy supply is only of limited importance. Besides, motivation would play a more important part for the moment of the termination of those exercises. Alternatively, intensities with very high anaerobic demands would shorten the exercise duration too much, so wouldn't so accurately reflect the demands specific to endurance-trained athletes. Moreover, statements on hormonal secretion could be limited if the exercise duration is too short, because changes of blood hormone levels would be influenced to a higher extent by shifts in the plasma volume and a decreased hepatic clearance.[12,13] Therefore, an exercise test performed at an intensity of 110% of the individual anaerobic threshold[14] and, thus, at about 10% above the maximal lactate steady state,[15] can be regarded as useful for an examination of exercise-induced hormonal concentrations during overtraining. In this context the exercise time until exhaustion during this test has turned out to be a sensitive single criterion for the diagnosis of overtraining.[16,17]

1.2 Hormonal Monitoring of Overtraining

Apart from routine measurements of substrates, such as lactate, ammonia and urea, etc., or enzyme activity (creatine kinase) in the blood, the question of a possible objective monitoring of exercise intensity and training by determinations of hormones has arisen in the last few years.[18-27] Endogenous hormones are essentially involved in exercise-induced acute or chronic adaptations and influence the regeneration phase through the modulation of anabolic and catabolic processes after exercise. Hormonal changes also seem to play an essential role with regard to the pathogenesis of overtraining syndrome[2,4,6,16-18,29] (table I). Table I provides a summary of studies in which the authors suppose or suspect an overtraining syndrome.

Recently with a prospective longitudinal study, we investigated the behaviour of anthropometric, ergometric, cardiopulmonary, metabolic, hormonal and immunological parameters, as well as the psychological profile of the same group of athletes both in 'normal' (non-overtrained) and short term overtrained states.[16,17] The duration of observation of approximately 1.5 years made it possible for each athlete to provide three to four intra-individual normal values to characterise his or her baseline. Overreaching or overtraining was induced through a pronounced increase in training intensity with repetitive sessions of highly intensive endurance training (corresponding to the range of the individual anaerobic threshold and above), and intensive interval training, or frequent competitions. This also corresponds to other studies,[6,10,11,31] in which an essential role is attributed to the intensity of physical exercise for the development of overtraining syndrome. In particular it is during prolonged exercise at intensities above the individual anaerobic threshold, which no longer allow a lactate steady state, that the metabolic and psychological demands increase considerably, as can be seen from the disproportionate increase in free plasma catecholamines.[23,27]

Other studies report overtraining syndrome caused by the sole means of drastic increase in training volume[29,32] which in one study has been connected with a limited intake of carbohydrate leading to reduced muscular glycogen stores.[32] The long period of observation increased the likelihood of other stress factors e.g. infectious diseases, frequent competitions and psychological distress, occurring, in addition to the training often involved in generating overtraining syndrome. The multifactorial genesis of overtraining syndrome was noted as early as 1923 when Parmenter pointed out that other triggering factors should be looked for irrespective of training.

2. Adrenaline (Epinephrine) and Noradrenaline (Norepinephrine)

2.1 Catecholamines and Anaerobic Threshold

The behaviour of the free catecholamines measured in plasma, indicates the sympatho-adrenergic activation with respect to exercise.[33-35] Catecholamines modulate metabolic and cardiocirculatory reactions and adaptations to physical and psycho-

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Table I. Studies involving blood hormone levels of athletes in a s[illegible]f overtraining syndrome or overreaching over a period of at least 1 week

Reference	Number of participants	Follow-up	Overtraining	Performance	Blood hormones	Mood profile
Barron et al.[28]	6 M marathon runners (mean best performance 2h 50 min) 4 in OT	4 months	Intensive training, frequent races during 8-14 weeks	Racing and training performance ↓	Cortisol-, ACTH-, STH- and PRL-response to insulin-induced hypoglycaemia ↓; normal response of LH, TSH and PRL to LHRH	Emotional and behavioural symptoms
Häkkinen et al.[20]	21 M physically active subjects (2 groups)	36 weeks	3 strength training sessions per week (2 protocols) during 24 weeks	Maximal strength during the first 20 weeks ↑, followed by a 4-week plateau phase (OT ?)	During the training phase: cortisol ↓ and testosterone shortly ↑ in one experimental group (n = 11); during the plateau and the following regeneration phase testosterone/cortisol and testosterone/SHBG correlate to the changes of maximum strength; LH, FSH, STH and SHBG unchanged	Not specified
Adlercreutz et al.[18]	M long-distance runners		'Very intense training programme during 1 week'	Not Specified	Free testosterone/cortisol <70% or $<0.35 \times 10^{-3}$	Not specified
Kindermann[6]	Mid-distance runners		Case reports	Maximal anaerobic lactic performance on treadmill (stepwise increasing and speed endurance) tests ↓	Maximal exercise-induced rise of free adrenaline (epinephrine) and noradrenaline (norepinephrine) ↓ (parasympathicotonic OT)	Enumeration of vegetative complaints (different in sympathetic and parasympathetic forms of OT)
Häkkinen et al.[21]	11 M weightlifters	1 year including 6 weeks of preparation for main competition	11% increase of training volume (kg) during 2 weeks	No significant changes in competitive results (OT?)	After intensive training: testosterone ↓, LH ↑; after the following normal and reduced training: testosterone unchanged; cortisol and LH ↓; the changes of testosterone/SHBG correlate with those of the competitive results	Not specified
Kirwan et al.[92]	12 M swimmers, 4 in OT $\dot{V}O_{2max}$ 58 ml/kg/min	24 days	110% increase in training distance at 94% $\dot{V}O_{2max}$ for 10 days	Swimming tests, training speed in 4 subjects ↓	Cortisol ↑, resting catecholamines (↑), in all 12 subjects	Subjective rating of training effort ↑
Kraemer et al.[31]	7 M,F physically active subjects $\dot{V}O_{2max}$ 50.4 ml/kg/min	10 weeks	3 longer sprint intervals and 3 maximal 30 min runs per week	$\dot{V}O_{2max}$ ↑ (!), (OT?)	Cortisol at rest and after maximal incremental treadmill test ↑; ACTH and β-endorphin after exercise ↓	Not specified
Fry et al.[10]	5 M elite soldiers in OT	15 days	2 intensive interval running sessions per day on treadmill during 10 days	Time to exhaustion on treadmill at 18 km/h, 1% gradient ↓	Testosterone, SHBG, cortisol, LH, FSH unchanged	Subjective complaints (questionnaire)
Lehmann et al.[29]	8 M mid- and long-distance runners $\dot{V}O_{2max}$ 66.8 ml/kg/min	5 weeks	Approx 100% increase of training volume over 3 weeks	Treadmill running speed at 4 mmol/L lactate unchanged, total running distance in incremental test ↓	Submaximal free noradrenaline ↑; adrenaline (↑); cortisol at rest (↓), after exercise ↓; (free) testosterone, insulin, STH, C-peptide, prolactin, LH, FSH, thyroid hormones, aldosterone, free catecholamines at rest and after exercise unchanged	Daily rating of 'well-being' (scale from 1-4) ↓
Hooper et al.[46]	14 M,F swimmers, 3 in OT	6 months	Approx 10% increase of training volume during tapering for 2-3 weeks before trials	Competitive results ↓	Resting noradrenaline before the trials ↑; adrenaline, cortisol unchanged	Level of fatigue > 5 (scale from 1-7) over 7 days without illness
Flynn et al.[78]	5 M swimmers $\dot{V}O_{2max}$ 58.6 ml/kg/min	21 weeks	2 weeks of 88% increase in training volume with approx. the same percentage of high intensive training	Performance in swimming tests ↓	(Free) testosterone ↓; cortisol unchanged	Global mood state impaired
Urhausen et al.[16,17]	17 M cyclists and triathletes, 16 in OT $\dot{V}O_{2max}$ 61.2 ± 7.5 ml/kg/min	19 months	Approx 180% increase of time of intensive training over at least 2 weeks in addition to individual stressors (competitions, short recovery after illness, social pressure)	Time to exhaustion at 110% of the individual anaerobic threshold on a cycle ergometer ↓; performance at anaerobic threshold and in all-out 10s/30s-test unchanged	Maximal exercise-induced rise of ACTH, STH, insulin ↓, cortisol (↓), submaximal noradrenaline ↑; all hormones including testosterone, SHBG, LH, FSH (at rest), β-endorphin and free adrenaline/noradrenaline (at rest and maximal) unchanged	Impaired mood profile at rest and subjective rating of effort during ergometry

Abbreviations and symbols: ACTH = corticotrophin; F = female; FSH = follicle stimulating hormone; LH = luteinising hormone; LHRH = luteinising hormone-releasing hormone; sec = second; SHBG = sex-hormone–binding globulin; M = male; OT = overtraining syndrome; PRL = prolactin; STH = somatotrophic hormone; TSH = thyroid stimulating hormone; (↑)(↓) = tendencial, not statistically significant.

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logical work. During incremental graded physical work, free adrenaline (epinephrine) and noradrenaline (norepinephrine) levels correlate with the exponential increase in lactate.[36-38] During prolonged exercise with intensities below the individual anaerobic threshold, the catecholamines increase in a time-dependent manner although the blood lactate level remains constant or even decrease.[27] If the exercise intensity exceeds the individual anaerobic threshold by 5% or more the lactate level exceeds the maximal lactate steady state, which leads to progressive acidosis.[15] At the same time, there is an overproportional time and intensity dependent rise in adrenaline and noradrenaline as a sign of an increased sympatho-adrenergic activation (fig. 1). This would mean that, in accordance to the concept of the so-called 'aerobic-anaerobic transition',[39] the range of the individual anaerobic threshold seems to indicate a real physiological transition corresponding to the maximal lactate steady state. Shorter high-intensity (anaerobic lactic) exercise is characterised by a higher catecholamine secretion and, in addition, show a comparatively higher adrenaline/noradrenaline ratio.[23,39-41]

During prolonged exercise at 85 to 100% of the individual anaerobic threshold, the levels of adrenaline and noradrenaline only demonstrate rather discrete differences.[27] This can be explained by the well-known variability of the individual hormonal levels[42,43] as well as by the duration of exercise. In earlier studies with greater differences between exercise intensities and longer exercise duration (90 to 100 minutes), the catecholamines stabilised at an intensity of 75% of the individual anaerobic threshold after the 50th minute, while, at an intensity of 100% of the individual anaerobic threshold, the levels continued to increase steadily.[23] Apart from thermoregulatory demands, the increase in catecholamine levels in the course of the prolonged exercise is to be attributed to a reduced intramuscular glycogen availability during training.[44,45]

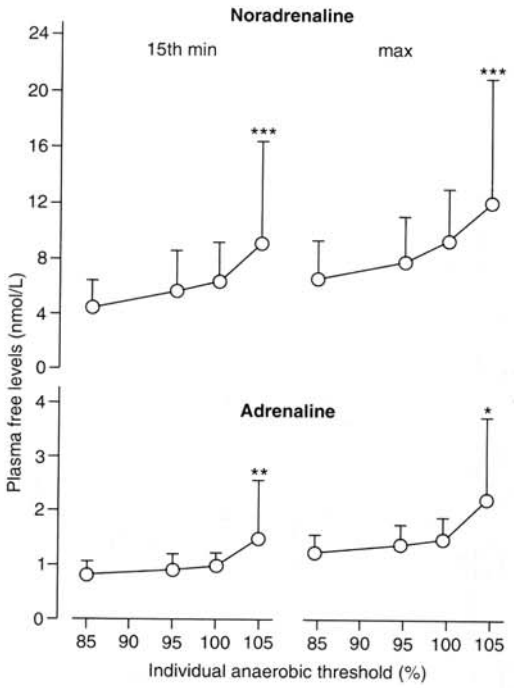


Fig. 1. Mean (± SD) free plasma adrenaline (epinephrine) and noradrenaline (norepinephrine) during 45 minutes of lasting endurance running on treadmill compared with the intensity of the individual anaerobic threshold in 14 endurance trained male runners (reproduced from Urhausen et al.[27], with permission). *Symbols:* * indicates p = 0.05; ** indicates p < 0.05; *** p < 0.01.

2.2 Catecholamines in Overtrained Athletes

In chronically overtrained runners (parasympathicotonic form) a disturbed glycolytic energy mobilisation with reduced maximal blood lactate levels has been described.[6] This has been characterised by lowered plasma values of free adrenaline and noradrenaline after maximal stepwise increasing and speed-endurance (exercise duration 1 to 2 minutes) treadmill tests in comparison to nonovertrained runners. In a retrospective study of three swimmers, increased noradrenaline values have also occasionally been reported at rest,[46] but this could not be confirmed by Kindermann[6] or by our own more recent studies and the maximal exercise test.[16,17] In more recent investigations, with unchanged maximal catecholamine concentrations, a partly sympathicotonic form rather than

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the (chronic) parasympathicotonic overtraining was observed,[4,16,17,29] thus the form of overtraining should be considered. During submaximal exercise, however, increased noradrenaline values were measured.

It can be speculated that there is a counterregulatory decrease in the sensitivity and density of the adrenoreceptors during overtraining on account of chronically increased catecholamine levels in training.[29,47,48] A compensatory increase in the submaximal noradrenaline level, caused by the impaired glycolysis in overtraining, is unlikely because of the nondiffering (only tendencially increased) values of adrenaline. A modified neuronal activity possibly plays a role through muscle receptors,[29] metabolites in active muscles could stimulate the adrenergic activity via afferent fibres during exercise.[50,51] A connection with the occasionally reported slightly increased submaximal oxygen uptake could also be possible.[17,52] It is also believed that catecholamines are less important to glycolysis and glycogenolysis during physical exercise than has generally been assumed,[4] because a blood lactate level of about 8 mmol/L could still be measured at low exercise intensity in patients with defect sympathic regulation (Shy-Drager-Syndrome).[53]

2.3 Catecholamines and Psychic Stress

In future, a starting point for measurement of overtraining could be the application of measurements of catecholamines in the assessment of sympatho-adrenergic activity of different training forms as well as in competition. Exercise forms with high nervous stress components are probably more likely to induce overtraining syndrome, especially if the adequate periods of recovery are neglected. This information does not necessarily result from blood lactate levels. During competition, badminton players showed similar adrenaline and noradrenaline values in comparison with highly intensive training, whereas lactate levels were about three times lower.[54] This reflects the pronounced psychological stress specific to competition, which has also to be taken into account for the periodisation of training. The assumption[33,55] cannot be generalised, that adrenaline reflects the mental stress and noradrenaline rather the physical one.[56] High psychological stress during physical exercise was followed by pronounced increases in noradrenaline.[56-58]

2.4 Methodological Prerequisites

Generally, the contribution of the measurement of free plasma catecholamines in the diagnosis of overtraining syndrome appears unclear or at least problematic at present. Intraindividual variability of catecholamine levels, which is increased when measured with different assays, has to be considered.[42] As far as monitoring of training by catecholamine determinations for the prevention or diagnosis of overtraining syndrome is concerned, however, it is essential that blood samples drawn at different times are also measured with different assays. The assessment of individual standard values is therefore problematic.

Compared with venous blood samples, measurement of capillary plasma catecholamines (e.g. from the earlobe) is advantageous, if the practicability and validity are considered, and this method can be used during training and competition. In addition, capillary catecholamines seem to be better indicators of the sympathic activity than the venous levels.[56,59] The capillary values are comparable to the corresponding results obtained by arterial samples.[60] The application of these so-called micro-methods, however, requires particular analytic demands, especially if the determination of lower adrenaline values is desired. At present radioenzymatic measurements are, with few currently tested exceptions, the methods of choice.[61,62] It is usually not possible to repeat measurement because of the small sample volumes (about 100µl plasma). Also, in comparison with other blood hormones and lactate, the half-life time in plasma of the free catecholamines is very short (about 2 minutes), so that the blood drawing has to be carried out as early and as fast as possible after exercise.

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3. Testosterone and Cortisol

Depending on the intensity and duration of preceding physical work, hormones with anabolic or catabolic properties, such as testosterone and cortisol, show quantitative changes signalling a catabolic state,[12,13,19,63,69] which can be reversed by appropriate regenerative measures[18,20-25,27,70] (fig. 2). This has led to interest in controlling the periodisation of training through determinations of blood levels of testosterone and cortisol ('hormonal monitoring of training') to prevent or even to diagnose overtraining. For the latter, Adlercreutz et al.[18] proposed a decrease in the ratio of (measured) free testosterone/cortisol >30% or $<0.35 \times 10^{-3}$. The biologically active free testosterone can either be measured directly or calculated by using the blood level of total testosterone and sex-hormone–binding globulin (SHBG).[18,70-74] The latter method seems to be preferable at least for routine application. In the Adlercreutz study,[18] the athletes had taken part for one week in a 'very intensive training programme'. Further descriptions on methods, study participants, complaints or objective performance data were not made.

3.1 Testosterone

3.1.1 Findings in Strength Athletes

Häkkinen et al.[20,21] observed a parallel development between the testosterone/cortisol ratio and the training-induced changes in strength in power-athletes for 36 weeks or 1 year, and also during a plateau phase of performance, which was interpreted as overtraining (see table I). An increase in strength, however, is not necessarily connected with a simultaneous increase in the testosterone/cortisol ratio.[75] Similar findings with regard to power athletes were published by this group.[19,76] It also seems that initially poorly-trained individuals show unchanged hormonal serum values during progressive strength-training.[77]

3.1.2 Findings in Endurance Athletes

Only a few useful findings concerning the behaviour of (free) testosterone or the testosterone/cortisol ratio in relation to objective parameters of performance in endurance athletes are available. Flynn et al.[78] observed a decrease in (free) testosterone connected with a simultaneously reduced swimming velocity and mood state after the training volume had been increased by 88% for two weeks. In prepubescent boys a correlation between indices of physical performance (endurance, strength and speed) and the increase in testosterone was described over one year.[79] After 6 months of physical training, as part of the military service, other investigators found an increase in testosterone,[80] which correlated with the maximal oxyg uptake.

3.1.3 Findings in Female Athletes

Female athletes also are reported to present lower testosterone values in comparison to untrained women[81,157] although few studies exist. With respect to the low physiological testosterone level (about 7 to 10% of male values) and a low number of study participants, an in principle comparable, but less close, correlation between the behaviour of blood testosterone and the training intensity was described in female rowers[25,82] and speed skaters.[73] A significant negative correlation between the free testosterone/cortisol ratio and the distance rowed has been found during a training camp.[82] As in a previously described group of men,[20,21] the individual total and free testosterone levels also correlated significantly with the changes of muscle cross-sectional area in fema students even during a short term (3 weeks), but intensive strength training period, however, without leading to significant changes of testosterone, SHBG or cortisol.[77,83] In addition, the authors found larger interindividual differences in the serum testosterone levels in comparison with the men and concluded that testosterone is an important indicator of trainability for muscle hypertrophy and/or strength development.

If the hormonal data of female athletes are taken into consideration, the influence of menstrual cycle or oral contraceptives mainly on blood levels of testosterone, cortisol and SHBG has to be considered.[82,84,85] However, the seasonal variations in

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the free testosterone/cortisol ratio did not differ among the users and nonusers of oral contraceptives in one long term study.[82]

While, in male athletes, the (free) testosterone/cortisol ratio was mainly determined by the behaviour of (free) testosterone,[25,26,86] in female athletes (n = 3), the variations of this ratio have been reported to be primarily determined by the behaviour of cortisol, which is supposed to behave in a more stable way even during the menstrual cycle.[74,82] Some authors[87] also consider that free testosterone remains relatively constant.

3.1.4 Overtraining Studies

In many training[86,88-91] and overtraining studies,[10,16,17,29,92] however, there were no essential changes in levels at rest of testosterone, cortisol and SHBG during overtraining nor was there a correlation with performance data. It is probable that the behaviour of (free) testosterone and cortisol is rather a physiological indicator of the current training load, but it does not necessarily indicate overtraining syndrome.[4,19,22,25,26,73,89,92,93] The testosterone/cortisol ratio is occasionally also overinterpreted. An increase of this ratio is, for example, interpreted as a 'good hormonal condition' for the next training period or the increase in the ratio (following a highly intensive training-induced decrease) measured after a regenerative training period is interpreted as a 'positive consequence of the total training'.[94]

At present, the endocrinological and basic ultracellular biochemical mechanisms are insufficiently understood. Other authors draw conclusions from changes of the testosterone/cortisol ratio for one season together with unchanged competition results from strength and endurance athletes on an insufficient intensity of training.[95] A decrease in the free testosterone/cortisol ratio >30% is occasionally described as an 'indicator of an insufficient regeneration'[73,89] without, however, presenting objective performance data. Although significantly reduced ratios were sometimes measured combined with a decline in performance during competition,[25,64] even a decrease in (free) testosterone/cortisol of >30%

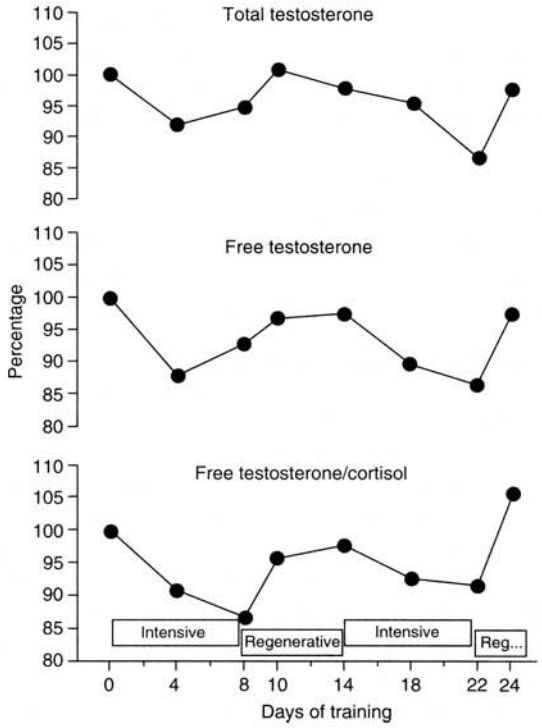


Fig. 2. Mean changes in serum total testosterone, free testosterone (as calculated by the total testosterone to sex-hormone-binding globulin ratio) and free testosterone/cortisol during several weeks, with strictly controlled (ergometric laboratory) intensive and regenerative training. The 7 endurance-trained male athletes were aged 25 ± 4 years. Maximum oxygen uptake ($\dot{V}O_{2max}$) was 60.8 ± 4.0 ml/min/kg. The initial value (day 0) had been set at '100'.

during intensive training or competition phases,[24,82,86,89,93] does not necessarily indicate overtraining syndrome. The transferability of the absolute low limit[18] of the free testosterone/cortisol ratio of 0.35×10^{-3} to other study conditions is also problematic, because the athletes' initial values characterising their 'normal' state frequently are already decreased by about 50%.[95-97] The statements of other investigators,[73,89] which differ by the factor 10^3 from the original above-mentioned ratio mentioned of the Finnish authors (both hormonal levels expressed in the same units) [e.g. Aldercreutz et al.[18] and Härkönen, personal communication] cause additional confusion.

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Moreover, the definition of a threshold with respect to the decrease in testosterone/cortisol as a sign of a 'hormonal overtraining' probably makes it necessary to consider individual differences which could also be sport-specific.[73,89] The influence of a disturbed energy balance could not often be discerned from the effect of physical exercise: it is known that an insufficient dietary intake can induce changes of peripheral hormones by decreasing the metabolic clearance of some steroids.[98]

3.1.5 Sex-Hormone–Binding Globulin

The evaluation of changes with regard to the carrier protein SHBG is unclear. An SHBG increase in response to a testosterone decrease lasting over a longer time is thought to suggest overtraining.[18,22] An increased level of the carrier protein somewhat protects the testosterone pool from a precipitate metabolic turnover. In contrast, low SHBG values are regarded as being a sign of an 'insufficient adaptation' to training loads.[99] This statement, however, was made without supporting performance data and is based on empiric observations. On the other hand, increased SHBG levels lead to a further decrease in the biologically active testosterone fraction, which could be interpreted to be an unfavourable shift of the 'anabolic/catabolic balance'.[99] An influence of diet on the SHBG level[63,100] with increased values during fasting and correspondingly decreased free testosterone fraction[98] is known. In most of the studies, SHBG showed no changes during overtraining (table I), so that total testosterone mainly parallels the changes of free testosterone.

3.1.6 Testosterone Levels During and After, Exercise and Training

The serum testosterone shows a biphasic behaviour under physical exercise: after exercise of shorter duration, testosterone increases in relation to intensity, volume and the muscle mass involved on account of a decreased hepatic and extra-hepatic clearance or stimulation of β-receptors in the testis.[36,66,72,101,103] From a duration of exercise of approximately three hours, the values decrease.[66,104] During the regeneration phase, decreased levels at rest were observed for hours to days e.g. after triathlon competitions,[68,105] but also after repeated anaerobic exercise.[30,67] As far as the frequently lower blood testosterone levels[96] in endurance athletes as compared with untrained individuals are concerned, it was criticised, that these values were almost always obtained retrospectively by means of cross-sectional studies, that there were wide differences among the study participants described as endurance-trained and that non–training-induced factors, such as diet, age, differences of the time of blood sampling (e.g. circadian as well as seasonal variations)[106] and, in particular, genetic influences remained unconsidered in most cases.[96]

It is assumed,[107,108] but partly denied,[109] that a transient suppressive effect on testicular luteinising hormone (LH) receptors as a result of the exercise-induced increase in corticotrophin (ACTH) and cortisol is the reason that the testosterone levels are decreased by endurance training or during overtraining. A direct influence on the testicular testosterone secretion could play a role, such as through the inhibition of an enzymatic step or a decreased sensitivity of the Leydig cells to stimulating hormones.[110,111] A cortisol-induced suppression of the testosterone secretion[112] is ruled out by some authors[109,113] because of unchanged cortisol levels at rest. Others assume a catecholamine-induced increase in the testicular vascular resistance,[114] a desensitising effect on LH receptors through reduced prolactin levels[115] or an increased metabolic rate of testosterone.[116,1 Many reports, however, suggest a central mechanism in the sense of a (supra) hypothalamo-pituitary regulation with disturbed gonadotrophin-releasing hormone (GnRH) release and LH pulsatility. For further discussion, see section 4.1.

3.1.7 Clinical Relevance

To date there have not been either any well-founded or rather contradictory results on the question of whether training or overtraining-induced decreased testosterone levels in men are really able to induce a biologically unfavourable effect. In runners, the training mileage has been found to correlate with sperm motility and density. At the

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same time, total and free testosterone were significantly lower in high mileage runners.[113] However, both the sperm quality and the testosterone and LH levels of the high mileage runners were still measured within the (low) normal range. The uncompromised sperm analysis in runners found by other investigators[118,119] could be influenced by methodological limitations.[113] In some cases, decreased testosterone values together with a subjective feeling of tiredness, a tendencially impaired spermatogenesis (n = 5) and a decreased libido are reported after an endurance training of only 14 days.[120] High intensive training loads during 5 days, however, did not lead to any changes of the above-mentioned parameters.[121] In another case, a very low testosterone level is discussed as the reason for an osteoporotic fracture in one walker.[122]

3.1.8 Metabolic Effects

To what extent decreased testosterone levels influence the energy-supplying processes cannot be definitely answered at present. Testosterone could play an important role to the muscle metabolism during the recovery phase in so far, as its anabolic effect does not only concern the protein synthesis,[123] but also seems to increase the ability of the muscle to refill its glycogen storage, which is emptied after physical exercise, through an increased activity of the muscle glycogen synthetase.[124,125] This would be very important to an adequate regeneration after prolonged exercise and during intensive training periods. While temporarily decreased testosterone levels seem to be only of minor importance, because the testosterone of the interstitium is primarily essential for the biological activity with regard to the target cells, this does not apply to testosterone levels that decreased for a prolonged period of time.[70] In this case, an anabolic deficit is assumed, which is characterised by a decreased counterregulatory effect at receptor level with respect to the catabolic effect of cortisol.[126]

Androgen pathways (besides others like growth hormone-related factors) significantly influence the exercise-induced muscle hypertrophy.[127] In animal experiments, an increased protein catabolism as the result of the activation of intracellular nonlysosomal proteinases after the administration of glucocorticoids could be proved.[128] Exhausting prolonged exercises have effects comparable to glucocorticoids. In experiments on isolated rat muscle, this effect could be reversed or prevented[128,129] through the administration of anabolic steroids or testosterone. However, the importance of testosterone for the endurance training–induced adaptation of muscle protein metabolism is sometimes questioned in other studies based on animal experiments.[130] It has also been shown that sensitivity to androgens was enhanced by a rapid increase in the number of receptors in exercised muscles.[131]

3.2 Cortisol

3.2.1 Overtraining Studies

Repeated determinations of cortisol have been recommended as tool for the diagnosis of overtraining. A parallel behaviour of the values at rest to training intensity[21,91,92,132] as well as increased levels during overtraining[28,31,90,133] are reported. Other investigators, however, did not find any increased cortisol values at rest during overtraining,[16,17,30,45,78] but even levels which were tendencially decreased[162] or, in spite of training reduction, 'unexpected' cortisol increases.[95] Higher cortisol levels are also interpreted as a long term adaptation to training in oligo-amenorrhoeal female skiers and orienteerers, suggesting a chronic catabolic state[134]. However, no training or performance data were presented.

Conversely, the maximal exercise-induced cortisol rise has been measured to be decreased during overtraining.[16,17,28,29] In contrast to that, an increased cortisol response under exercise is described as a training-induced adaptation,[88,135] which, however, is not confirmed by all investigators.[136] Nevertheless, it can be assumed that different training forms (aerobic or anaerobic) also lead to different hormonal adaptations.[31]

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3.2.2 Cortisol During and After Exercise

The exercise-induced cortisol increase depends on the duration and intensity of physical exercise.[12,13,64,65,137-139] A significant increase in the blood cortisol level usually requires a duration of exercise of more than 20 minutes with at least 60% of the $\dot{V}O_{2max}$ and is primarily the consequence of a higher secretion rate.[140,141] During the post-exercise phase, cortisol usually decreases rapidly and, within hours, reaches the initial value again.[13,67,68] Apart from physical strain and psychological factors (e.g. mental stress before starting the effort) diet probably also plays a role for the behaviour of cortisol[132] through a modified metabolic clearance.[98]

3.2.3 Metabolic Effects

The cortisol increase is especially important for the energy supply when exercise duration is prolonged, since cortisol prevents the re-esterification of fatty acids released by the catecholamine-induced lipolysis.[12,139,142] During the post-exercise phase, a participation of cortisol in the muscular glycogen resynthesis is speculated.[143] Also, a transference of glycogen depots to the liver at the expense of the muscle stores was reported.[123] Through the decrease in glucoplastic amino acids with increased gluconeogenesis, however, cortisol has a protein-catabolic effect on the skeletal muscle,[123,144] which is probably of importance during recovery.

In summary, although testosterone and cortisol are the most frequently investigated hormones as a measure of overtraining, hormonal monitoring of training does not yet seem to be suitable, as is the case for the lactate diagnostic.

4. Pituitary Hormones

4.1 Gonadotrophins (Luteinising Hormone and Follicle-Stimulating Hormone)

After acute physical exercise, LH usually decreases and the LH response to GnRH remains uninfluenced.[28,145,146] Some investigators even found an increased pituitary-gonadal response after strenuous exercise.[147] The follicle-stimulating hormone (FSH) usually does not show significant exercise-induced changes. During very intensive training phases, a reduced pulsatile LH secretion and response was found in marathon runners and athletes after the administration of GnRH.[97,148,149] However, the influence of an insufficient energy intake on the pulsatile LH release should also be considered.[150] This would suggest an impaired hypothalamic-pituitary mechanism. The present studies, however, used single or serial blood sampling over a few hours and not 24-hour serial samplings.

It is uncertain whether isolated measurements hormonal levels in peripheral blood are sensitive enough to investigate these complex mechanisms. A decrease in testosterone levels after 6-months of training or in high-mileage runners was not connected with an impaired LH secretion.[109,113] Findings in overtrained (eumenorrhoeic) female athletes (n = 4) with significantly reduced pulsatile GnRH release (which depends on regulation by neurotransmitters such as β-endorphin and dopamine) and LH secretion[151,152] also indicate a central mechanism for the development of overtraining syndrome. In female athletes these changes are related on menstrual cycle irregularities.[3] Some authors claim that training-induced amenorrhoea may be one manifestation of overtraining in female athletes.[153] Impairments of bone density in amenorrheic athletes leading to an increased risk of stress fractures are well described[154,155] and even seem partly irreversible.[156] Caloric deficits or eating disorders may play an additional important role.[157,158]

4.2 Corticotrophin (ACTH)

Despite speculations on an essential participation of hypothalamo-pituitary hormonal changes in the pathogenesis of overtraining, only few results with respect to athletes, who were actually in a state of overtraining are available. We observed an impaired exercise-induced increase in corticotrophin, somatotrophic hormone (STH) and insulin after an exhausting short-endurance test at 110% of the individual anaerobic threshold[16,17] (fig. 3). An in-

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teresting parallel to our results exists with the findings of Barron et al.[28] The latter were able to prove a reduced increase in corticotrophin, growth hormone and cortisol in response to an insulin-induced hypoglycaemia in 4 overtrained marathon runners, while there was an adequate response of the pituitary hormones after the administration of the LH-releasing hormone (LHRH). It was concluded that a hypothalamic dysfunction exists in the sense of a central depression in overtraining. In animal experiments,[159] a transient impaired pituitary function with lower corticotrophin values after repetitive stimulation by electrical shocks was also described, and interpreted as a reaction of self-protection of the organism.[69] This was ascribed to a negative feedback through cortisol or a depletion of the pituitary corticotrophin pool. The corticotrophin secretion depends on the intensity and duration of the exercise and is stimulated by different stressors, especially by decreasing levels of blood glucose.[12,123,137,160,161] It primarily occurs as a response to the corticotrophin-releasing hormone (CRH), but is also influenced by catecholamines.[144] Higher corticotrophin levels were found in trained as compared with untrained women during exercise, which is rated as an indication of possibly increased pituitary or hypothalamic sensitivity by the investigators.[162]

In other studies, which described lowered corticotrophin values after a 3-month endurance training,[163] peripheral mechanisms are speculated to be involved, such as a change of the intracellular homeostasis influenced by chemoreceptors.[164] The feeling of heavy legs as one typical complaint with overtraining and which occurs despite low lactate values,[17,165] could be connected with such a modified sensitivity of chemoreceptors. In athletes during overtraining Kraemer et al.[31] also reported an impaired exercise-induced rise of corticotrophin. However, they found increased cortisol values. In our study,[16,17] the exercise-induced increase of cortisol after the stress test was reduced, but less than the one of corticotrophin. A different, enhanced sensitivity of the adrenal cortex[88] does not seem to be the cause, since it has been found to be reduced[166] or unchanged[3] in overtrained horses.

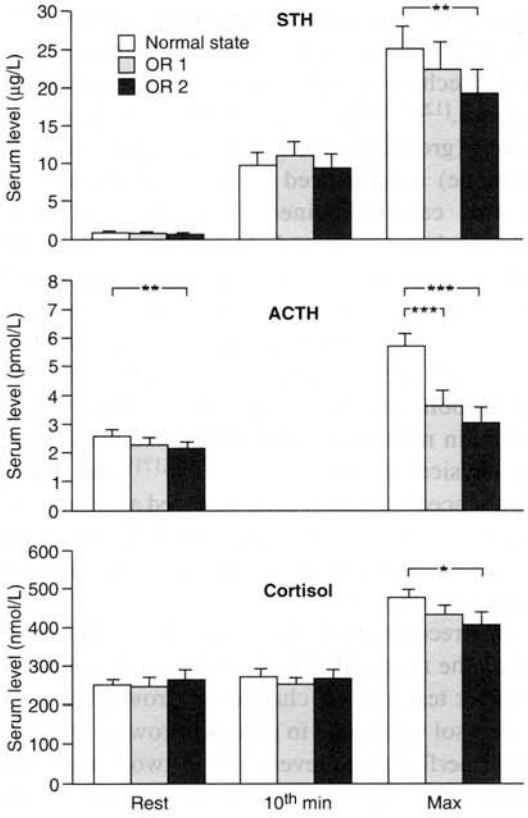


Fig. 3. Serum levels of somatotrophic hormone (STH), corticotrophin and cortisol before, during (tenth minute) and after (maximal values) an exhaustive bicycle endurance test at 110% of the individual anaerobic threshold (corresponding to 83% $\dot{V}O_{2max}$). Means were ±SE, n = 17 endurance trained male triathletes and cyclists. *Abbreviations and symbols:* OR 1 = overreaching; OR 2 = severe overreaching and/or overtraining; * indicates p = 0.06; ** indicates p < 0.05; *** indicates p < 0.01.

4.3 Somatotrophic Hormone

The interpretation of STH levels makes it necessary to consider that most of the metabolic effects of overtraining cannot necessarily be related on the blood levels, since the effect is induced by STH-dependent serum factors – somatomedines e.g. the insulin-like growth factor 1 (IGF-1).[167,168] The exercise-induced increase in muscle

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mass is at least partly independent of STH.[12,69] The STH release is a hypothalamic controlled pulsatile mechanism with numerous sleep-induced secretions.[12,123] The secretion of hypothalamic hormones (growth hormone-releasing and inhibiting hormone) is influenced by neurotransmitters (including catecholamines), hormones (including adrenal hormones and endogenous opoioids), amino acids as well as physiological stimuli (e.g. stress, exercise, hypoglycaemia and sleep).[145,169,170] So the exercise-induced STH release itself seems to be at least partly responsible for anabolic effects of exercise.[168] The increase occurs in relation to the duration and intensity of the physical exercise[12,65,67,142,167,171] and is also pronounced especially after repeated anaerobic exercise forms.[142,172] At present, the IGF-1 response to exercise is not clearly established and controversially reported.[168]

In a recently published study, a correlation between the maximal performance in a 7-minute ergometer test and the changes of growth hormone or cortisol was found in groups of rowers with different performance levels.[173] In two studies significant correlations between the maximal oxygen uptake and blood levels of IGF-1 are reported.[174,175] In overtrained athletes we measured an impairment of the maximal exercise-induced STH response, while training enhances the pulsatile release or the secretion of STH under physical exercise.[44,171,176] In a blood sample taken immediately after a stepwise increasing treadmill ergometry, other authors measured unchanged STH levels during overtraining.[29] The intensity characterising the 'stress test' we used as the ergometric parameter for the objective diagnosis of overtraining[16,17] had been initially fixed at 110% of the individual anaerobic threshold (approx. 80% of $\dot{V}O_{2max}$ in the participants of this study) to obtain a comparable exercise-induced rise in the growth hormone as it was the case with the insulin-induced hypoglycaemia[170] used in the above-cited study of Barron et al.[28] Interestingly, the spontaneous nocturnal secretion of the growth hormone during highly intensive training phases was reported to be impaired,[177] but increased in normally-trained athletes.[178]

4.4 β-Endorphin

The behaviour of β-endorphin with regard to shorter aerobic exercise, partly presents conflicting data in the literature. On the one hand, there is an approximately 5-fold increase after a 30-minute exercise with 80% of the $\dot{V}O_{2max}$[179] and, on the other hand, there are unchanged β-endorphin values after a 20-minute run with 80% of the maximal heart rate.[145] A certain threshold intensity (approximately corresponding to the range of anaerobic threshold) has to be exceeded in order to measure significant β-endorphin responses to exercise.[180,181] With endurance exercise at an intensity in the range of the individual anaerobic threshold, the exercise duration usually has to exceed 60 minutes before significantly increasing blood levels of β-endorphin can be measured.[181-183] Studies suggest a relationship between the increase of β-endorphin and the extent of anaerobic lactic energy supply (or acidosis tolerance) as well as an enhancement of hormone secretion from training.[181,182] Endogenous opioids inhibit the GnRH release.[145] In some studies, a decreased β-endorphin response after physical exercise[31,184] and in the stimulation test[184] was reported in overtrained athletes. The same applies to female rats.[185] However, under β-blockade, there is an earlier increase in β-endorphin connected with a simultaneously higher subjective perception of exertion and faster muscular fatigue. The β-endorphin level was not decreased at the maximal exercise level, although the maximal exercise intensity level was decreased.[186]

4.5 Prolactin

The response of prolactin (PRL) to exercise reveals large interindividual differences. It depends partly on the intracellular glucose availability.[12,163] The exercise-induced rise in PRL only seems to be significant in better-trained individuals[12,135] or at higher exercise intensities (>90% $\dot{V}O_{2max}$ in untrained people). As a result of training,

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the release of PRL is reported to be enhanced or starts at lower exercise intensities,[163,187] but could not be confirmed in all studies.[164] The exercise response may be associated with anaerobic lactic contribution,[188] which would be interesting in view of the hypothesis of anaerobic lactic exercise forms being one important trigger mechanism for inducing overtraining syndrome. Overtraining syndrome induced by an excessive increase in training volume did not lead to any changes of resting and exercise-induced PRL values,[29] which was also true for those in high mileage runners.[113] After a strenuous 5-day ranger training (an army combat course) combined with sleep deprivation and caloric deficit,[63] however, the levels at rest were reduced (as well as testosterone and estradiol). The meaning of these changes remains unclear. Elevated PRL values inhibit the GnRH release through an enhanced dopaminergic tone.[189,190] At present, there are no recommendations for using PRL measurements in the monitoring of training or diagnosis of overtraining.

5. Other Hormones

5.1 Insulin

Hardly any well-founded data with respect to athletes in intensive training periods or even in a state of overtraining exist for the other hormones. In our own study, we found a significantly lowered maximal exercise-induced level of insulin without altered blood glucose nor correlation to the duration of exercise.[16,17] In one case, Kuipers and Keizer[3] report on an athlete with parasympathicotonic overtraining showing a hypoglycaemic state under physical exercise, but without reference to levels of insulin or other hormones. Even in the earliest proposals with regard to the attempt of an objective diagnostic of overtraining syndrome, a decreased blood sugar level has already been mentioned as a sign of a glycogen depletion in overtraining.[191] During prolonged physical work the insulin level decreases, because the catecholamine increase inhibits the insulin secretion.[12,65,171,192] This behaviour favours glucose homeostasis with increased glucose availability for the central nervous system.[139] At the onset of exercise, the adrenergous inhibition of the insulin secretion disappears: the blood glucose level leads to the reincrease in insulin, which enhances the muscular glycogen repletion during the post-exercise phase. In addition, insulin accelerates the muscular growth both by means of somatomedines and insulin receptors, probably through decreased proteolysis.[101] Training has been shown to increase the insulin response after exhausting endurance exercise.[171] This is not contradictory to the insulin-sparing effect of a higher level of physical activity, which applies to submaximal rather than to maximal exercise intensity.[193] At the same glucose levels, the maximal insulin increase, however, was lower in trained than in untrained people after glucagon stimulation.[41] In contrast to our results, another study reports unchanged maximal insulin levels in overtrained athletes, but only one single measurement was carried out directly after the termination of exercise in the step-test.[29]

In support of the hypothesis of a (supra) hypothalamo-pituitary axis impaired in overtraining, Elias,[145] report an enhanced hypothalamic opioid tone as a result of low insulin levels[194] This could result in a depressed GnRH pulsatility and subsequent gonadotrophin secretion, as seen in insulin-deficient patients.[195] High insulin levels attenuate the central increase of neuropeptides.[196]

5.2 Thyroid-Stimulating Hormones and Estradiol/Progesterone

With respect to measurements of the blood levels of thyroid hormones or thyroid-stimulating hormone (TSH), no findings that would suggest an applicability in the monitoring of training or diagnosis of overtraining[29] exist at present. After a stressful combat course of 5 days, the secretion of thyroxine and triiodothyronine decreased, while TSH was depressed.[63]

As far as changes of estradiol or progesterone levels in female athletes are concerned, there are no reports on changes applicable to the control of

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periodisation of training or even the diagnosis of overtraining. The methodological difficulties of a hormonal monitoring of training by measuring these parameters varying during the menstruation cycle are evident. During exercise, additional problems would arise from lower sex-steroids binding affinities of plasma proteins because of increased body temperature and competitive interactions between androgens and estrogens.[87]

6. Special Conditions

6.1 Diet

Differences in caloric intake or diet composition are well known to affect physical performance, but also the anabolic-catabolic hormonal status and/or response to exercise. A low fibre and complex carbohydrate, high fat and protein intake is associated with low SHBG and high plasma sex hormone levels.[100] After 7 months of a reduced fat and high carbohydrate diet in male ice hockey players (free) testosterone, cortisol and LH increased in comparison to controls.[197] However, short-term modifications in diet (3 days) were not associated with concomitant changes in the concentration of anabolic and catabolic hormones (cortisol, SHBG, IGF-1) despite of changes in physical performance data.[198] Seven days of a high (70%) carbohydrate diet diminished the exercise-induced increase of STH after caloric restriction in comparison to a control group with 55% carbohydrate intake.[199] An inadequate caloric intake, or eating disorders, are also well-known to impair central hormonal regulations including a reduced pulsatile LH secretion.[150,200] Furthermore, an imbalance in caloric intake has been described to decrease the metabolic clearance of steroid hormones.[98]

6.2 Hormonal Treatment or Manipulation

One additional influence on blood hormonal concentrations could be the intake of banned substances, especially anabolic hormones. In fact, one of the major reasons for use of testosterone and anabolic steroids, growth hormone, clenbuterol, erythropoeitin and releasing factors during training seems to be the supposed prevention, or treatment, of (hormonal) overtraining. The evidence from earlier studies on hormonal status and responses in athletes may be misleading, especially because of the use of such hormones. The athletes are not always willing to reveal the use of hormones and the researchers are not always informed about it. So it is possible that some results and conclusions are misleading. It has been shown that exogenous testosterone depresses endogenous secretion of testosterone as well as the releasing hormones including corticotrophin.[201,202] After withdrawal sustained steroid administration the pituitary function can be affected leading to long-lasting impairment of testicular endocrine function[201-204] in terms of a hypogonadotrophic hypogonadism. The concentration of growth hormone and estradiol are also reported to increase by administration of testosterone or aromatisable anabolic-androgenic steroids in men.[202]

6.3 Altitude

An acute exposure to altitude conditions can also significantly influence blood hormonal concentrations. During exercise at altitude several factors should be considered if comparing hormonal and performance results: degree of hypoxia (moderate or high altitude conditions), climate conditions (temperature, humidity), duration of exposure and acclimatisation, individual experience of altitude training, plasma volume changes (haemoconcentration) as well as the most often reduced absolute exercise intensity. However, if considering similar relative workloads (percentage of maximal oxygen uptake), acute moderate hypoxia seems to not affect metabolic and hormonal responses to short bouts of exercise (four 5 min exercise bouts).[205] A more detailed presentation of the complex hormonal changes during exposure to altitude thus would exceed the limits of this review.[206]

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7. Pathophysiological Considerations

7.1 Energy Supply

At present there are only few findings concerning the role of anabolic and catabolic hormones under discussion during the recovery phase following physical exercise. The impaired secretion of centrally or peripherally acting hormones during overtraining contrasts with the usually increased hormonal levels induced by training.[207] A decreased exercise-induced rise of pituitary hormones, cortisol and insulin as well as lower resting levels of testosterone possibly affects the resynthesis of protein and glycogen during the regeneration phase after physical exercise.[12,69,70,125] The lower respiratory quotient repeatedly described in overtraining[17,52,208] probably indicates a shift of the energy-supplying processes in favour of an increased fat and a decreased carbohydrate utilisation. In this context, the submaximally higher noradrenaline levels with their suppressive effect on the insulin secretion are possibly also important.[12] In hypophysectomised rats, the refilling of muscular glycogen decreased during the first 4 hours after exhaustive exercise, while, after 24 to 48 hours, no essential differences could be proved any longer.[209]

It is possible, that during overtraining peripheral and central mechanisms have a synergetic effect (fig. 4). It can be postulated that repetitive stress leads to an altered neuroendocrine regulation, thus possibly inhibiting the pulsatile hypothalamic release of GnRH through the CRH and/or opioid pathways. CRH inhibits GnRH, LH and growth hormone release.[145,146] The (supra) hypothalamo-pituitary-adrenal axis can also be influenced by peripheral stimuli, e.g. the accumulation of lactic acid, changes in pH or activation of muscular chemoreceptors.[145]

The repeated exposure to stress hormones (particularly cortisol) increasing during prolonged intensive exercises could lead to a suppression of the hypothalamo-pituitary axis with subsequent impaired release of corticotrophin and cortisol consecutively.[69,210] In chronically stressed rats, a restricted muscular Na^+,K^+-ATPase activity in connection with reduced cortisol levels was repeatedly discussed, suggesting the necessity of an intact pituitary-adrenal axis for an adequate function of the muscular Na^+,K^+-pump.[69] The Na^+,K^+-pump is responsible for maintaining the membrane potential and counteracts the loss of potential caused by the extracellular potassium increase with decreased excitability. Training enhances the activity of the Na^+,K^+-pump[211,212] and increases the membrane potential in animals and humans.[212,213] The Na^+,K^+-pump is primarily, but not exclusively, activated by β_2-receptors.[214,215] An ion shift with increased extracellular potassium level is also discussed as a possible cause for the occurrence of similar complaints during exercise under β-blockade,[215,216] as they are observed in overtraining. It is possible that the muscular fatigue is at least partly caused by such a mechanism.[217-220] To some extent, this could be an explanation for the muscular complaints in overtraining.

7.2 Amino Acid Dysbalance Theory

Some authors[221] speculate that as a result of the reduced muscular glycogen availability in overtraining, there are higher blood levels of tryptophan in comparison with branched chain amino acids on account of a higher muscular utilisation of branched chain amino acids and/or a shifted binding of tryptophan to albumin in connection with the increased occurrence of free fatty acids. The transformation into the neurotransmitter serotonin (5-hydroxytryptamine; 5-HT) in the brain is assumed to cause central fatigue. Differences in the run time of rats could be attributed to changes in the brain serotonin activity, perhaps in interaction with subsequent modifications of the brain dopamine level.[222] Increased muscle and liver glycogen contents were found compared with controls, which could be in keeping with our assumption of an impaired intramuscular glycolytic mobilisation in overtrained athletes. However, an enhanced serotonin activity has been found to elevate the plasma levels of hormones resulting from the hypothalamo-pituitary-adrenocortical axis and

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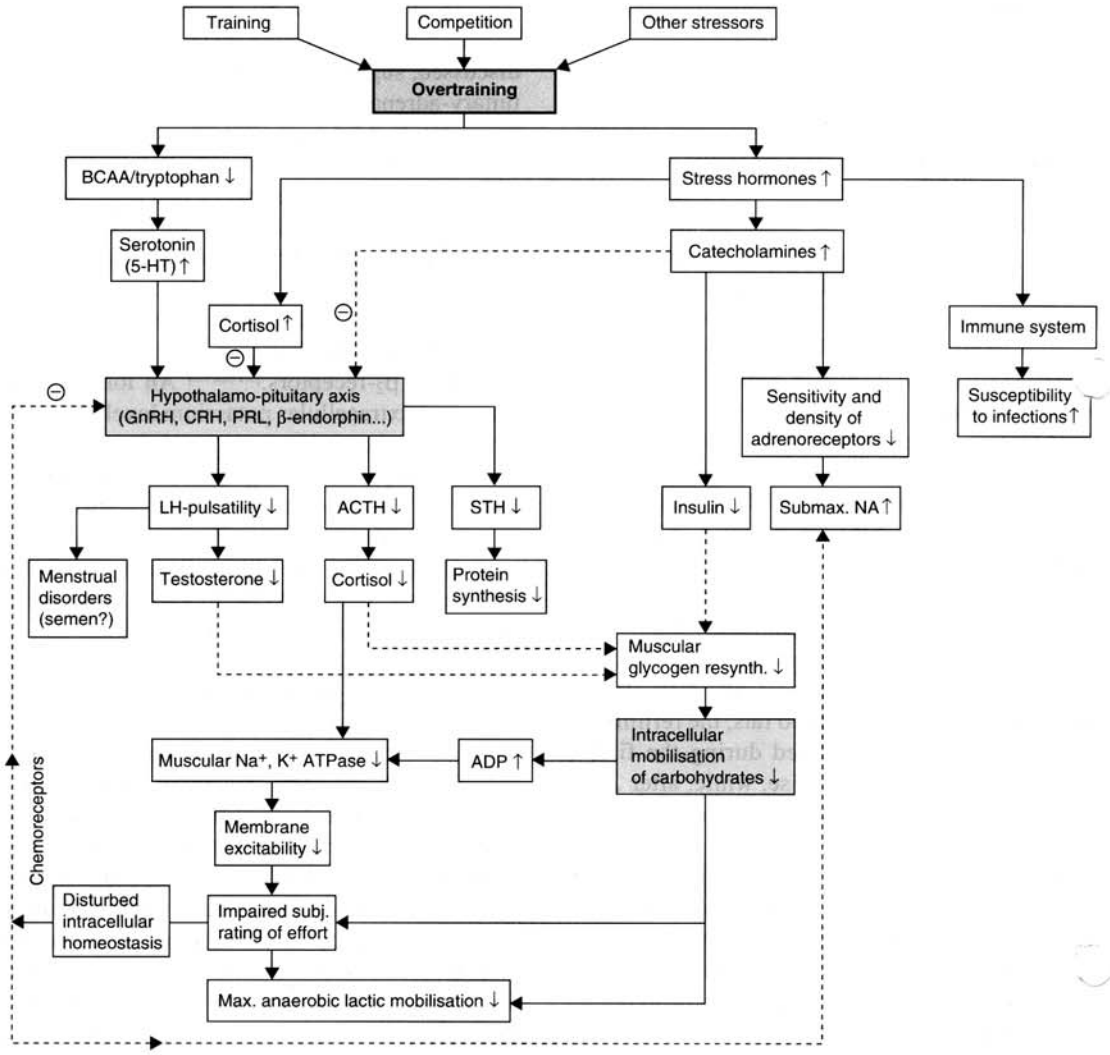


Fig. 4. Postulated neuroendocrine mechanisms in overtraining. *Abbreviations and symbols:* ACTH = corticotrophin; ADP = adenosine diphosphate; BCAA = branched chain amino acid; CRH = corticotrophin releasing hormone; GnRH = gonadotrophin releasing hormone; LH = luteinising hormone; PRL = prolactin; NA = noradrenaline (norepinephrine); STH = somatotrophic hormone; subj. = subjective; submax = submaximal; ↑ = increases; ↓ = decreases.

sympathoadrenal system.[223,224] Although these data result from rats at rest and, thus, cannot be simply transferred to exercising humans, the so-called 'amino-acid dysbalance theory' could be understandable in the context of acute fatigue in prolonged exercise rather than in chronically overtrained athletes, who are characterised by reduced maximal levels of pituitary hormones and free catecholamines. In addition, intracerebral injections of serotonin induce sleep,[221] which fits with the

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fatigue during prolonged exercise but contrasts with the sleep disorders of many athletes during overtraining.

7.3 Immunofunctions

Reports on an increased susceptibility to infectious diseases of overtrained athletes lead to the consideration of links between hormones and the immune system. Hormones like adrenaline and cortisol induce a leucocytosis and modulate the functional properties of immune cells[225] Furthermore the exercise-induced acute-phase response consecutively leads to a hormonal response of, for example, corticotrophin and cortisol, to inhibit local exercise-induced inflammatory processes.[226] Current data about the alteration of the immune system by overload training are incomplete. But results indicate a (transient) modification of the regulation of the T and B cell functions, which at the moment might be interpreted as modulated cell surface antigens indicating enhanced activity and yet unclear proliferative response of T cells as well as a suppressed ability of B cells to produce immunoglobulins.[10,227-229] A further interesting finding is the decreased glutamine level in the blood after 6 weeks exhaustive running training, if the essential role for the energy supply of immune cells is regarded.[230] It might be speculated, if repetitive hormonal stressors lead to a reduced glutamine synthesis in the muscles, which conversely could have an influence on the mono- and lymphocytes functional response. The increased susceptibility for infections of overtrained athletes[231,232] will likely find an explanation in a fine but complex regulation of both closely interacting hormonal and immune systems.

8. Future Research

8.1 Intraindividual Follow-up Studies

Future research dealing with hormonal measurements in training athletes mainly results from two areas. First, systematic measurements of selected hormonal parameters will improve the current knowledge of positive adaptive mechanisms following training, and also a better understanding of overtraining. Interactions with neuroendocrine, peripheral metabolic and immunological pathways are of special interest. Secondly, the application in the sense of a hormonal monitoring of training could be possible, in which the repeated measurements of hormonal blood levels allow information on the current tolerance of effort and, thus, the periodisation of training through the planning of high intensive or recovery periods (table II).

Statements with regard to overtraining syndrome are only possible through studies involving a sufficient number of athletes actually a state of overtraining syndrome, but at least in the state of overreaching. It seems not to be suitable to simply transfer results obtained during intensive phases of (overload) training. Since the diagnosis of overtraining syndrome, however, remains a clinical diagnosis through the exclusion of underlying organic illness, studies with an adequate methodology can only be carried out, if an effective cooperation between skilled experts in endocrinology and physicians specialised in sports medicine and working in direct contact to athletes is guaranteed.

8.2 Central and Peripheral Regulation

Future basic research should include the (supra) hypothalamic level e.g. by considering releasing

Table II. Problems and prerequisites of hormonal monitoring of training

Lack of reference values: considerable individual differences
Limited knowledge about the regulation (supra-hypothalamic level) and peripheral modulation (receptor level)
Methodological limitations
Relatively high inter- and intra-assay variability; so measurements in the same assay hardly practicable
Results must be quickly available
Need of relatively large sample volumes
Expensive
Circadian/pulsatile rhythms
Influences of external factors e.g. nutrition, altitude, hormonal therapy or manipulation, psychic stress. . .
Different plasma half-lives
Female athletes: physiologically low testosterone levels, menstrual cyclic variations

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hormones (CRH could represent an important key),[145] stimulation tests or measurements of electroencephalic activity. A more exact recording of the activity of hypothalamic-pituitary axis, however, requires serial determinations of pulsatile secreted hormones. Regarding the peripheral regulation of hormonal action, the receptor level also remains relatively unexplored although it modulates the effect of the measured blood levels. To the best of our knowledge, there have not been any systematic studies on muscular hormone receptors in athletes during overtraining. In high-intensity training including frequent anaerobic exercises, an upregulation of β-receptor density on lymphocytes at rest has been described as opposed to the effect of endurance training.[233] These examinations, however, inevitably involve considerable methodological and technical efforts.

8.3 Methodological Challenge

The systematic use of hormonal measurements in the monitoring of the intensity of training is currently somewhat restricted by the rather high intra- and inter-assay variability of the laboratory methods in comparison with the absolute hormonal changes measured during overtraining (table 2). In contrast to the substrates and enzymes, which, so far, have been measured in the monitoring of training as a matter of routine, the determinations of hormones also remain problematic because of their dependence on circadian and/or pulsatile rhythms, psychological factors and others. This requires carefully standardised sampling conditions as well as individual comparative data. As far as measurements during and after exercise are concerned, the varying half-life of the different hormones is also to be taken into account. The influence of psychological factors, however, could also be advantageous, because psychological stress factors often play an important role with regard to the development of the overtraining syndrome.

At present there are no standard values which indicate when recovery is complete, and the next training stimulus can be set again or when there is an insufficient recovery with the danger of overreaching. For this reason, systematic longitudinal studies with highly trained athletes are necessary to measure hormones regularly related to clinical examinations, psychological data and objective parameters of performance under standardised conditions. These studies naturally involve many efforts concerning the methodological aspects as well as the recruitment of subjects and are also more difficult when including female athletes, because of the more complex cyclic hormonal variations. The development of micro-methods allowing hormonal measurements from capillary blood samples promise to be very helpful, so that athletes will accept more frequent testing. Under these conditions, it is expected, that hormone measurements in blood will, in future, represent an essential link in the concept of psycho-endocrine regulation of metabolic as well as immunological reactions, and adaptations to physical exercise.

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Correspondence and reprints: *Axel Urhausen* MD, PhD, Institute of Sports and Preventive Medicine, University of Saarland, D-66041 Saarbrücken, Germany.

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